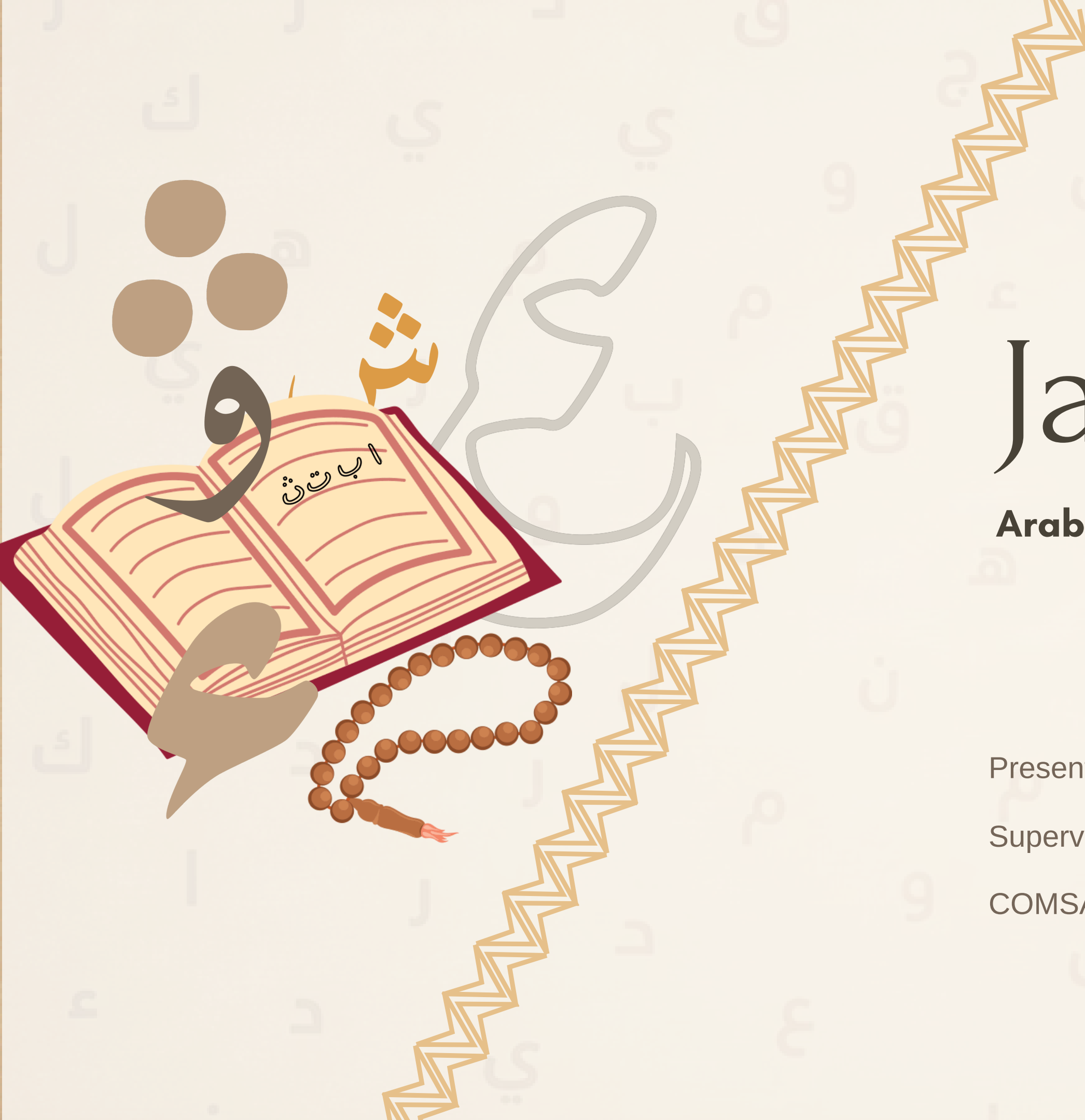




بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ



Jawhar

جوہر

Arabic at its Core

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The Problem

Existing Tools

Duolingo, Rosetta Stone — designed for conversational modern languages

CAMeL Tools, Farasa, Anki — powerful but siloed and not integrated into a unified learning experience

Insufficient coverage

Classical and Quranic Arabic differ considerably from Modern Standard Arabic and are not supported by mainstream language-learning tools.

Global comprehension gap

Over 1.8 billion Muslims worldwide lack access to tools that enable understanding of the Quran in its original language.

Tool mismatch

Platforms such as Duolingo and Rosetta Stone target conversational modern languages and do not address Quranic grammar depth (sarf, nahw).

Fragmented capabilities

No single platform currently unifies NLP morphological analysis, syntactic parsing, AI tutoring and spaced repetition into one cohesive Quranic learning experience.

Our Solution

One unified platform purpose-built for Quranic Arabic comprehension.

Delivered as a cross-platform application on Android and iOS via Kotlin Multiplatform.

NLP-Powered Analysis

Automatic morphological breakdown (Sarf) via CAMEL Tools and syntactic parsing (Nahw) via Farasa on any Quranic verse.

AI Conversational Tutor

A Gemini Flash-powered tutor that answers learner questions about grammar, meaning, and structure in real time.

Adaptive Vocabulary & Paths

A spaced repetition system for Quranic vocabulary combined with learning paths that adjust based on individual performance.

Objectives



1 Cross-platform application

Build a Kotlin Multiplatform app for Android, iOS and Desktop to ensure broad accessibility.

2 Morphological analysis (Sarf)

Deliver root extraction and verb-conjugation analysis using CAMEL Tools.

3 Syntactic analysis (Nahw)

Provide grammatical parsing and i'rab using Farasa's syntactic modules.

4 AI conversational tutor

Integrate Gemini Flash to provide interactive tutoring, explanations and targeted feedback.

5 Spaced Repetition System

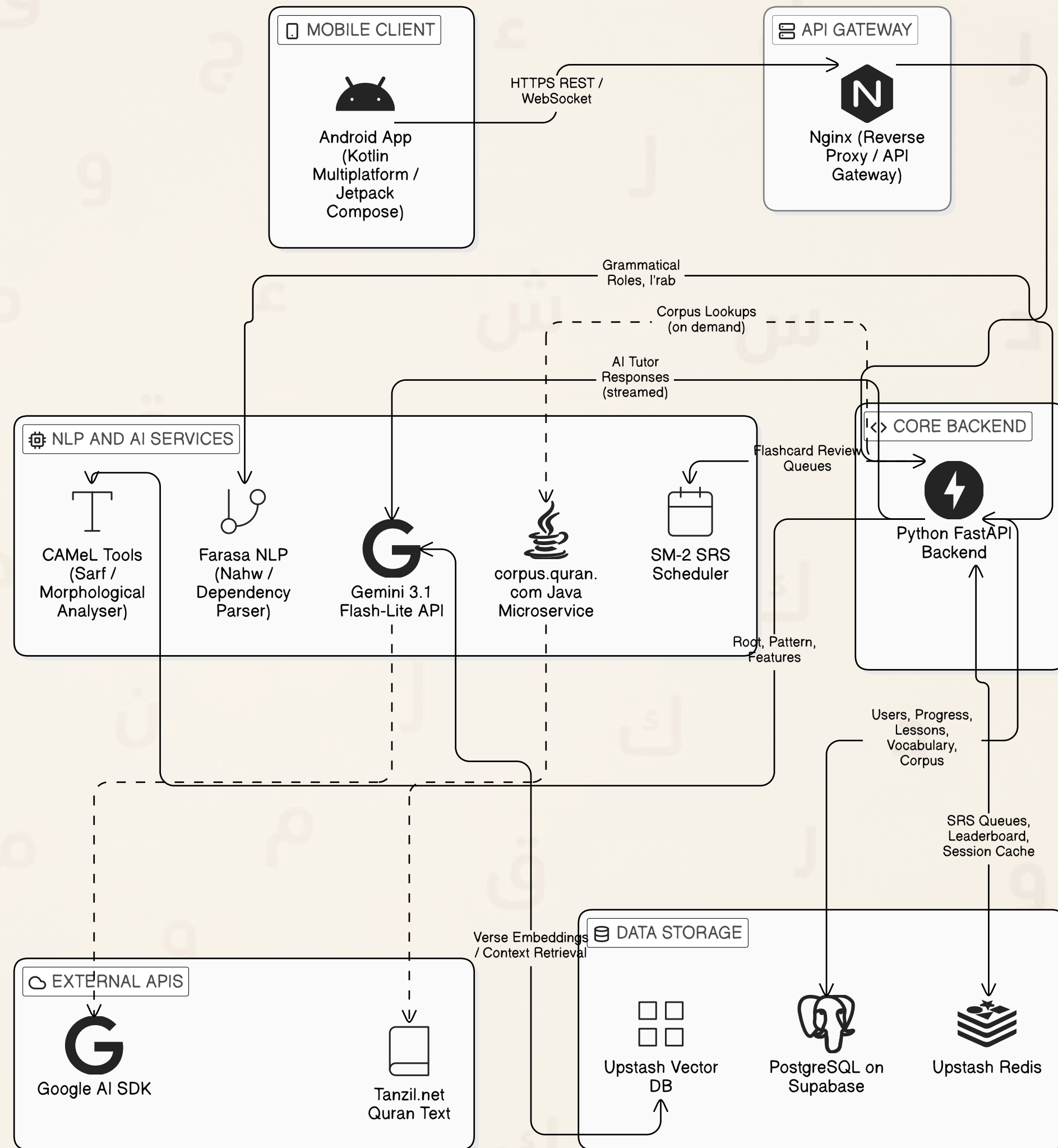
Implement SRS to optimise vocabulary and morphology retention across study sessions.

6 Adaptive learning paths

Personalise curricula dynamically according to learner performance and diagnostics.



System Architecture



Technology Stack

Component	Technology
Frontend	Kotlin Multiplatform (Android, iOS)
NLP — Morphology	CAMeL Tools (Python)
NLP — Syntax	Farasa (Python)
Quranic Corpus Microservice	corpus.quran.com API (Java)
AI Model	Gemini Flash (interactive tutoring)
Database	PostgreSQL
Cache	Redis
Infrastructure	Docker, Nginx, GitHub Actions



Project Timeline

Phase 1

Research &
Architecture

Phase 4-5

(M5–6) Backend &
Corpus API

Phase 8-9

(M9–10) AI Tutor &
SRS Engine

Phase 2-3

(M3–4) NLP
Pipeline

Phase 5-6

(M7–8) Frontend
Development



Questions & Discussion



Thanks You!

Presented by: Meher Ali & Hammad Ali